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Orioles DC	Royals DC	Astros DC
Rays DC	Tigers DC	Athletics DC
Red Sox DC	Twins DC	Mariners DC
Yankees DC	White Sox DC	Rangers DC

NL East	NL Central	NL West
Braves DC	Brewers DC	D-backs DC
Marlins DC	Cardinals DC	Dodgers DC
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Orioles	Royals	Astros
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20262025

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BAL CHW ATH
 BOS CLE HOU
 NYY DET LAA
 TBR KCR SEA
 TOR MIN TEX

NL

ATL CHC ARI
 MIA CIN COL
 NYM MIL LAD
 PHI PIT SDP
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[The Beginners Guide to the Positional Adjustment](#)

by [Neil Weinberg](#)

February 27, 2015

Getting newcomers on board with Wins Above Replacement has a number of challenges, but the way we measure and evaluate defense is typically one of the biggest sticking points. Getting an open-minded person to believe in wOBA instead of average and RBI isn't that difficult. Getting someone to accept that there's more to base running than the number of stolen bases is pretty easy. Convincing them that it's useful to compare players to replacement level is a bit harder, but nothing really compares to the questions people have about defense.

There's good reason for this. Again, a thoughtful person can see the flaws in using errors or fielding percentage, but it's harder to sell the merits of runs saved metrics for a number of reasons. If you want a little more information on how we measure defense and why we do it that way, check out our [beginner's guide to measuring defense](#). Today, we're going to consider a corollary to the actual measurement of defense which is the [positional adjustment](#).

In essence, the positional adjustment is a mathematical correction to account for the fact that different positions are more challenging than others. An average center fielder is worse than an average first baseman. That's a pretty easy thing to accept. A position might be more challenging because there's a higher volume of plays or a higher degree of difficulty. For our purposes, it's not important to get too far into the weeds here. Shortstop is more challenging than first base or left field, and we can leave it at that for a moment.

Now we have to think about the way we currently measure defense, and we measure defense by comparing players to league average at their position. So if you're familiar with Ultimate Zone Rating (UZR) or Defensive Runs Saved (DRS), you know that those numbers show up as runs above or below average at that position. So a shortstop who is five runs better than the average shortstop gets a +5 and a left fielder who is five runs better than the average left fielder gets a +5.

We compare players to their positional averages because we want to base player evaluations on a similar set of plays. It would be silly to compare the conversion rate of a third baseman against that of a right fielder because not only do they field a different number of balls, they're just a totally different kind of batted ball. As a result, when you look at the initial defensive statistic, you're looking at only how that player compares to their fellow position-mates.

And when it comes to overall player evaluation and WAR, we need some way to balance out those differences in position. We need to find a way to account for the difference between the positions. The positional adjustment does just that, and we'll get to when the specific numbers come from shortly.

In general, we want to add runs for players who play tough positions and subtract runs for players who play easier positions to account for the fact that average at one does not equal average at the other in terms of total run prevention. For example, for a shortstop, the adjustment is +7.5 runs per 1,458 innings (9 innings * 162 games) or a full season. For a left fielder, it's -7.5 runs per full season.

The difference between an average left fielder and an average shortstop is about 15 runs per season. That's a sizable gap. The positional adjustment is a method for putting a bunch of different positions onto one uniform scale so that when you add the adjustment to a player's runs saved to get what we call "DEF," you can compare players no matter their position.

The actual numbers we use are based on some calculations that were done about a decade ago that used the performance of players who moved positions, and it's certainly reasonable to suggest that those numbers have changed as the game has changed. The DH adjustment might be too negative and the catcher adjustment might be a bit too large. There's lots of room to disagree on the precise decimals and if you're so inclined, I'd invite you to come up with a more accurate rendering of the numbers.

But the important thing is that we need a positional adjustment. We calculate defense based on position average but the position averages need to be corrected so that an average catcher and an average first baseman don't look the same. The positional adjustment does this for us. It might make sense to come up with a better version of the numbers we use, but adjusting for position is a vital piece of overall player analysis.

[Stats to Avoid: Batting Average](#)

[Interpreting Playoff Odds and Projected Standings](#)

Neil Weinberg is the Site Educator at FanGraphs and can be found writing enthusiastically about the Detroit Tigers at New English D. Follow and interact with him on Twitter [@NeilWeinberg44](#).

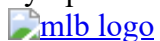
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Mitchel Lichtman

All UZR (ultimate zone rating) calculations are provided courtesy of Mitchel Lichtman.

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